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RESEARCH ARTICLE

Heterogenous Social Media Analysis for Efficient Deep Learning Fake-Profile Identification

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ABSTRACT The surge in e-scams attributed to an estimated 30% of fake social media accounts has highlighted the urgent need to identify such fraudulent profiles. Since the current model cannot handle multi-model networks, an attempt has been made to solve the real-time problems. This study introduced a cutting-edge deep-transfer learning model that streamlines fake-profile detection through a comprehensive analysis of diverse social media data samples. Our model gathers a wide range of data from various social media platforms, such as posts, likes, comments, multimedia content, user activity, login behaviors, etc. Each data type is individually processed to detect suspicious patterns synonymous with fake accounts—for instance, discrepancies like male profiles predominantly posting about or using images of females. Similarly, audio signals undergo 1D Fourier, Cosine, Convolutional, Gabor, and Wavelet Transforms. In contrast, image and video data are processed with their 2D counterparts. Text data is transformed using Word2Vec, aiding our binary Convolutional Neural Network (bCNN) to distinguish between genuine and fake profiles. Feature optimization is handled by the Grey Wolf Optimizer (GWO) for 2D data and the Elephant Herding Optimizer (EHO) for 1D data, ensuring minimal feature redundancy. Separate 1D CNN classifiers, then classify the refined features to pinpoint fake profiles. The results from these classifiers are amalgamated through a boosting mechanism. Our results reveal an 8.3% increase in accuracy, 5.9% in precision, and 6.5% in recall compared to conventional methods. Testing on initial login and signup processes further validated our model's effectiveness, achieving a remarkable 93.5% accuracy in detecting illegitimate account activities, effectively mitigating cold-start challenges.

INDEX TERMS Social, media, fake, profile, audio, image, text, video, multidomain, GWO, EHO, CNN, multimodal, cold start, issues.

I. INTRODUCTION

The meteoric rise and ubiquity of big data platforms, especially social media, have ushered in a new era of challenges, notably identity theft. The attractiveness of these platforms has not gone unnoticed by spammers and con artists, leading to a surge in cybercrime activities like spamming, which primarily inundates users with unsolicited emails. The manifestations of these threats on Social Media Platforms (SMPs) vary, but their presence is undeniable.

A profound understanding of the intricacies of human interpersonal relationships can offer invaluable insights into

user behavior, preferences, and conversational tendencies. While instrumental in enhancing the quality of products and services, such insights may simultaneously be weaponized against unsuspecting users. An illustrative scenario is manipulating online discourse by actors unbeknownst to other participants, thereby altering the conversation's trajectory.

The allure of SMPs lies in their provision of anonymity, which ironically becomes their Achilles' heel. The veil of anonymity emboldens users to craft fictitious profiles, causing distress to unsuspecting victims. Activities spanning rumor-mongering, cyberbullying, and the deliberate dissemination of false information, as was evident during events like Hurricane Sandy in the US, are a testament to the evil use of these platforms. The inherent competitive nature of SMPs,

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TABLE 1. Comparative analysis of existing methods.

Reference	Year	Proposed Method	Strengths	Limitations
[1]	2023	DeepFakeNet	High accuracy in detecting deep fake images	Requires extensive training data
[2]	2023	ProfileAuth	Efficient with multimodal data inputs	Struggles with newly created fake profiles
[3]	2023	SocioGuard	Incorporates user behavior for detection	It can be bypassed with sophisticated bots
[4]	2022	FakeDetectRNN	Uses RNNs to detect temporal patterns in fake profiles	Computational overhead
[5]	2023	GraphFakeNet	Utilizes graph-based relationships for detection	Not effective for isolated fake accounts
[6].	2022	DeepProfileScan	Utilizes GANs to counter-generate fake profiles	Vulnerable to adversarial attacks
[7]	2023	SocioGAN	Uses adversarial training to refine detection	GANs require significant computational resources
[8]	2022	Profile2Vec	Embeds profile features into vectors for clustering	Less effective on sparse datasets
[9]	2023	FakeGuard	Real-time detection suitable for live platforms	Dependent on regular model updates
[10]	2022	SocialSVM	SVM-based method optimized for large-scale networks	Struggles with non-linear fake generation patterns

exemplified by metrics like “likes” or “followings,” further exacerbates the issue, prompting users to employ overt and covert tactics to gain an edge.

Further complicating the landscape are bots and cyborgs, automated entities with varying degrees of human intervention. Differentiating between a genuine user and these entities poses a considerable challenge. For instance, despite being fictitious, “Sybil” accounts originate not from real individuals but from systems, while trolls represent fabricated human accounts designed with malicious intent.

The challenges mentioned in Table 1 underscore the importance of establishing robust mechanisms to discern genuine profiles from fake ones. While considerable research has gone into detecting bot-generated profiles, identifying fake human profiles remains relatively uncharted territory. There’s a glaring inconsistency in the definition of what constitutes a ‘false identity,’ but it’s apparent that humans present a complex challenge with their nuanced behaviors.

Our recent foray into this domain, involving an analysis of Twitter’s features to detect impersonation, revealed that conventional metrics like friend and follower counts, while accessible through Application Programming Interfaces (APIs), fall short of accurately pinpointing fabricated identities & scenarios.

Consequently, the limitations inherent in the current models, exacerbated by their inability to scale or adapt to the real-time dynamics of SMPs, have paved the way for innovative solutions. In Section III, this work introduces a pioneering deep-transfer learning model aimed at addressing these shortcomings by leveraging diverse social media analytics.

Subsequent sections delve into the rigorous validation of this model across multifaceted criteria, culminating in conclusive observations and forward-looking recommendations.

A. MOTIVATION AND CONTRIBUTIONS

While offering an avenue for global connectivity and information sharing, the digital revolution has concurrently birthed a dark underbelly of deception. With an alarming estimate suggesting that nearly one-third of all social media accounts are fake, the digital realm faces an increasing threat from e-scams and misinformation campaigns. As many people rely on social media for news, communication, and even business, these fraudulent entities undermine trust, sabotage genuine interactions, and paves the way for cybercrimes. Furthermore, the existing models aiming to mitigate this issue often fall short due to their inefficacy, incapacity to handle multimodal data, or overwhelming complexity that hampers real-time application. As social media’s role in personal, professional, and societal spheres continues to grow, an urgent need exists for an advanced solution that ensures user safety, trust, and platform integrity levels.

B. CONTRIBUTIONS

Deep-Transfer Learning Model: Introduced an innovative model that harnesses the potential of deep-transfer learning. This approach offers improved accuracy and adapts to diverse and heterogenous social media data types, ensuring versatility for different scenarios.

Multimodal Data Analysis: Our model is not constrained to one or two types of inputs. Instead, it embraces a holistic

approach, analyzing posts, comments, multimedia components, user activity, and even login patterns, comprehensively scrutinizing potential fake profiles.

Advanced Feature Processing: By employing techniques such as 1D/2D Fourier, Cosine, and Convolutional Transforms for audio and visual data, alongside Word2Vec for textual content, our methodology ensures that no nuance of fraudulent behavior escapes the detection process.

Optimized Feature Redundancy: We've integrated the Grey Wolf Optimizer (GWO) and Elephant Herding Optimizer (EHO) to eliminate redundant features, ensuring the model's decisions are based on the data samples' most relevant and distinct characteristics.

Enhanced Classification Accuracy: Through a combination of separate 1D CNN classifiers and a subsequent boosting classifier, our model outperforms traditional methods by significant margins in accuracy, precision, and recall metrics.

Real-World Application: Beyond theoretical efficacy, our model was rigorously tested in real-time scenarios such as initial logins and signups, ensuring its practical relevance and potential for immediate integration into existing systems.

Thus, this paper carves out a pioneering pathway in the battle against fake social media profiles, providing a robust, efficient, and scalable solution tailored to the multifaceted nature of today's digital platforms.

II. DESIGN OF AN EFFICIENT DEEP LEARNING MODEL FOR IMPROVING FAKE-PROFILE DETECTION VIA HETEROGENOUS SOCIAL MEDIA ANALYSIS

As per the review of existing fake profile detection techniques, it can be observed that these models either showcase low efficiency or cannot be scaled for multimodal networks. Moreover, the complexity of these models is also high, which limits their applicability to real-time social media interfaces. To overcome these issues, this section discusses the design of a novel deep-transfer learning model for improving fake-profile detection via heterogeneous social media analysis. As per the model flow in Figure 1, it can be observed that the model initially collects multimodal inputs from different social media interfaces. These include posts, comments, like patterns, image components, video components, audio components, page-activity patterns, profile viewing patterns, and login-logout patterns.

Each of these patterns is individually processed to identify fake-profile behaviors. For instance, if fake profiles are created by 'males,' the profile photo will usually be of a 'female,' and all posts will probably be about females. Such patterns are detected by representing the collected components into multidomain feature sets. For audio signals, 1D Fourier Transforms, 1D Cosine Transforms, 1D Convolutional Transforms, and 1D Gabor and Wavelet Transforms are used, while for image and video signals, these transforms are represented in two dimensions. Text signals are represented via Word2Vec, which assists the binary Convolutional Neural Network (bCNN) model in classifying genuine and fake profiles. The 2D features are processed by a Grey Wolf

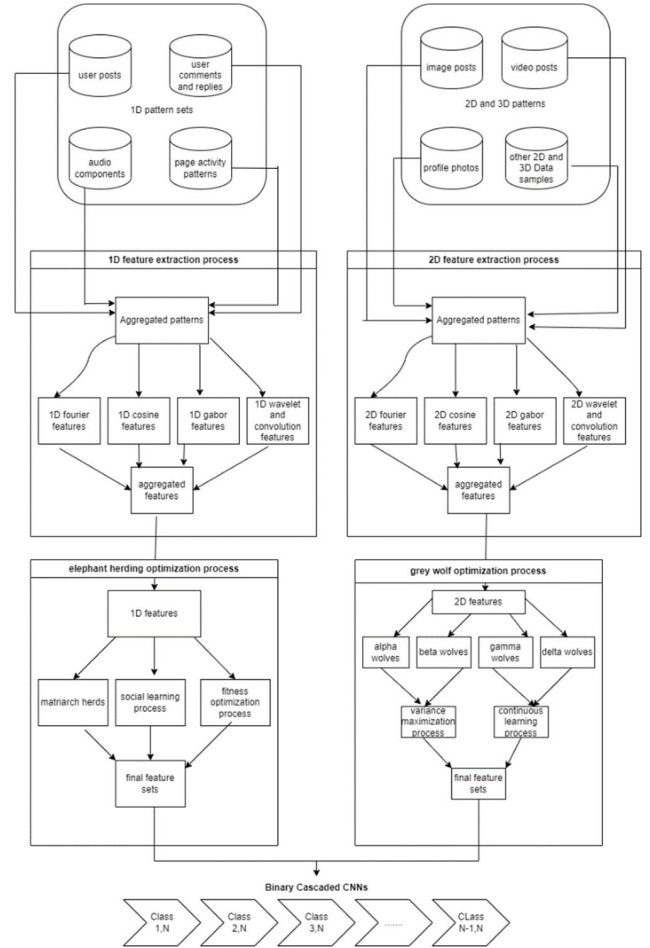


FIGURE 1. The overall flow of the proposed fake-profile detection process.

Optimizer (GWO), while the 1D features, including Word2Vec, are processed by an Elephant Herding Optimizer (EHO), which assists in reducing feature redundancies. Both the feature sets are classified via different 1D CNN classifiers, which assists in identifying fake profiles. To perform these tasks, this section segregates the proposed model design into multiple sub-modules, each described in different sub-sections of this text.

A. PROCESSING OF TEXT POSTS

All text posts are initially converted into 1D Feature Vectors via a Word2Vec-based model, which assists in representing individual words into numerical sets. To perform this operation, a dictionary of positive and negative words is used to obtain the polarity score of each word in the posted text. This is done via equation 1, where W_i , W_n , and W_p are input, negative, and positive words from the dictionary data samples, while P_{out} represents output polarity levels.

$$P_{out} = \frac{\sum_{i=1}^N |W_i == W_p| - \sum_{i=1}^N |W_i == W_n|}{N} \quad (1)$$

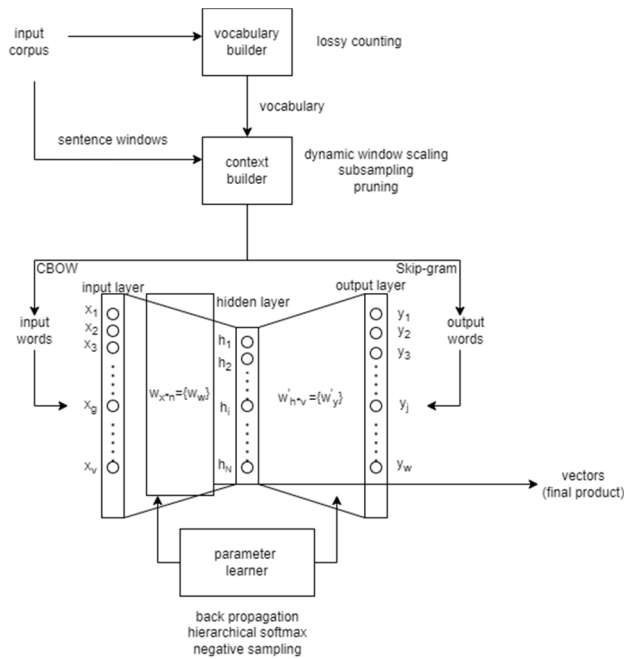


FIGURE 2. The overall flow of Word2Vec for extraction of text features.

where N represents the total input words. This polarity score, along with input corpus, is given to a Word2Vec Model, which can be observed from Figure 2 as follows, Word2Vec is a technique for sentiment analysis that may be used to create tools such as a thesaurus, context detector, and continuous bag of words (CBOW) engine. Examining the various sections of the document will disclose this reality. The program constructs a vocabulary by identifying “active words” in a given text. The context creator accepts the language and employs it to construct an appropriate environment. The context builder block provides word pairs to assist in locating word choices with related words. With the CBOW engine and skip gramme models having access to these variations, the feature extraction process may be conducted more effectively. These features are subsequently sent to the first step of a two-stage neural network, which converts the input word pairs into emotions. As an output of the overall sentiment analysis, this context-aware model may be utilized to identify the sentiment associated with each word combination. This engine is precise when labeling emotions and events, but training for real-time sample sets requires significant effort.

B. PROCESSING OF OTHER 1D FEATURES

Once text features are converted into numerical sets, other 1D Features, including login patterns, posting patterns, and page activity patterns, are converted into multidomain features. This is done to extract patterns from fake and normal user behaviors. To perform this task, Fourier Features (for Frequency Analysis), Cosine Features (for Entropy Analysis), Gabor Patterns (for Spatial Analysis), Wavelet Patterns (for enhanced Spatial Analysis), and Convolutional Features

(for Window-based Analysis) are extracted via equations, 2 through 8 as follows,

$$DFT_i = \sum_{j=1}^{N_f} x_j \times \left[\cos\left(\frac{2 \times \pi \times i \times j}{N_f}\right) - \sqrt{-1} \times \sin\left(\frac{2 \times \pi \times i \times j}{N_f}\right) \right] \quad (2)$$

where x represents the 1D patterns, N_f represents the total number of feature patterns extracted during behavioral analysis.

$$DCT_i = \frac{1}{\sqrt{2 \times N_f}} \times x_i \sum_{j=1}^{N_f} x_j \times \cos\left[\frac{\sqrt{-1} \times (2 \times i + 1) \times \pi}{2 \times N_f}\right] \quad (3)$$

$$Conv_{out_i} = \sum_{a=-\frac{m}{2}}^{\frac{m}{2}} x(i-a) \times LReLU\left(\frac{m+2a}{2}\right) \quad (4)$$

where m and a represent dimensions for different window and stride components, while LReLU is a Leaky Rectilinear Unit, which is used to activate negative feature sets via equation 5,

$$LReLU(x) = l_a \times x, \quad \text{when } x < 0, \quad \text{else } LReLU(x) = x \quad (5)$$

where, l_a An activation scaling constant converts negative features into positive feature sets.

$$G(x, y)_s = e^{\frac{-x^2 + \partial^2 \times y'^2}{2 \times \theta^2}} \times \cos\left(2 \times \frac{\pi}{\lambda} \times x''\right) \quad (6)$$

where x and y represent spatial dimensions of the input signal components, while ∂, θ and λ different angular and wavelength components are used to extract Gabor features.

$$W_a = \frac{x_i + x_{i+1}}{2} \quad (7)$$

$$W_d = \frac{x_i - x_{i+1}}{2} \quad (8)$$

where, W_a and W_d Represents approximate and detailed wavelet components. These components might contain redundancies, reduced via an EHO process, and discussed in this text's next sub-section.

C. SELECTION OF 1D FEATURES VIA EHO PROCESS

The extracted 1D Features are selected by an Elephant Herding Optimizer (EHO) process that assists in maximizing variance levels. This is done to reduce the delay needed for classification while improving the accuracy of fake-profile detection with multidomain features. This model works as per the following process,

- A set of NE Elephant Herds are generated as per the following operations,
- Select N stochastic features via equation 9,

$$N = STOCH(LE \times N(1D)_f, N(1D)_f) \quad (9)$$

where LE is an Elephant Learning Rate, a total number of extracted 1D features, and $STOCH$ is a stochastic number generation process.

- For each of the extracted features, an Elephant Herd fitness is calculated via equation 10,

$$fe = \sqrt{\frac{\left(\sum_{i=1}^N \left(x_i - \sum_{j=1}^N \frac{x_j}{N}\right)^2\right)}{N+1}} \quad (10)$$

- Once all Elephant Herds are generated, then an iterative fitness threshold is evaluated via Equation 11

$$f_{th} = \frac{1}{NE} \sum_{i=1}^{NE} fe_i \times LE \quad (11)$$

- Elephant Herds $fe > f_{th}$ are directly passed through to the next set of iterations, while other Herds learn from the 'Matriarch' Herd as follows,
 - A herd with maximum fitness is marked as a 'Matriarch' Herd
 - Features of all Herds with $fe \leq f_{th}$ are stochastically replaced with features of the 'Matriarch' Herd
 - In this case, $NH \times LE$ features are stochastically replaced, where NH is the total number of features extracted by the current Herds set.
- This process is repeated for NI iterations, and Herd with maximum fitness is selected to identify highly variant feature sets.

Once all iterations are completed, the selected features are classified via a binary cascaded CNN, described in section II-F of this text.

D. PROCESSING OF 2D AND 3D FEATURES

Once 1D features are extracted, 2D Features, including images, profile photos, and videos, are converted into multidomain features. This is done to extract patterns from fake and normal user behaviors. To perform this task, 2D Fourier Features are evaluated via Equation 12. Cosine Features are evaluated via Equation 13; Convolutional Features are extracted via Equations 14 and 15. Gabor Patterns are evaluated via equation 6; Wavelet Patterns are evaluated via equations 16, 17, 18, and 19 as follows,

$$DFT(r, c) = \sum_{i=1}^R \sum_{k=1}^C I(r, c) \times \left[\cos\left(\frac{2 \times \pi \times i \times j}{R \times C}\right) - \sqrt{-1} \times \sin\left(\frac{2 \times \pi \times k \times j}{R \times C}\right) \right] \quad (12)$$

where I represents the 2D patterns, while R, C are the size of 2D feature patterns extracted during multimedia analysis.

$$DCT(r, c) = \frac{1}{2 \times R \times C} \sum_{i=1}^R \sum_{k=1}^C I(r, c) \times \cos\left(\frac{\sqrt{-1} \times (2 \times i + 1) \times \pi}{2 \times R \times C}\right)$$

$$\times \cos\left(\frac{\sqrt{-1} \times (2 \times k + 1) \times \pi}{2 \times R \times C}\right) \quad (13)$$

$$Conv(i, j) = \sum_{a=-\frac{m}{2}}^{\frac{m}{2}} \sum_{b=-\frac{n}{2}}^{\frac{n}{2}} I(i-a, j-b) \times LReLU\left(\frac{m+2a}{2}, \frac{n+2b}{2}\right) \quad (14)$$

where m, n are window dimensions, a, b are stride sizes, while $LReLU(x, y)$ is represented via equation 15,

$$LReLU(x, y) = l_a \times I(x, y), \quad \text{when } I(x, y) < 0, \\ \text{else } LReLU(x, y) = I(x, y) \quad (15)$$

where l_a is an activation scaling constant used to convert negative features into positive feature sets.

$$W(A) = \frac{I(x, y) + I(x, y+1)}{2} \quad (16)$$

$$W(H) = \frac{I(x, y) + I(x+1, y)}{2} \quad (17)$$

$$W(V) = \frac{I(x, y) + I(x+1, y+1)}{2} \quad (18)$$

$$W(D) = \frac{I(x, y) - I(x, y+1)}{2} \quad (19)$$

where $W(A)$, $W(H)$, $W(V)$ and $W(D)$ represents approximate, horizontal, Vertical, and diagonal wavelet components. Similarly, components are estimated for 3D image and video sets. These components might contain redundancies, reduced via an efficient GWO process, and discussed in this text's next sub-section.

E. SELECTION OF 2D AND 3D FEATURES VIA GWO PROCESS

The extracted 2D Features are selected by an efficient Grey Wolf Optimization (GWO) process that assists in maximizing variance levels. This is done to reduce the delay needed for classification while improving the accuracy of fake-profile detection with 2D multidomain features. This model works as per the following process,

- A set of NW Wolves are generated as per the following operations,

- Select N stochastic features via equation 20,

$$N = STOCH(Max(LW) \times N(2D)_f, N(2D)_f) \quad (20)$$

LW is an augmented learning rate for individual Wolves, while $N(2D)_f$ is the total number of extracted 2D features.

- For each of the extracted features, an augmented Wolf fitness is calculated via equation 21,

$$fw = \sqrt{\frac{\left(\sum_{i=1}^N \left(x_i - \sum_{j=1}^N \frac{x_j}{N}\right)^2\right)}{N+1}} \quad (21)$$

- Once all Wolves are generated, then an iterative fitness threshold is evaluated via equation 22,

$$f_{th} = \frac{1}{NW} \sum_{i=1}^{NW} fw_i \times LW_i \quad (22)$$

- Based on this threshold, Wolves are marked as follows,
 - Wolf is Marked as 'Alpha' if

$$fw > 2f_{th} \quad (23)$$

- Else, Wolf is Marked as 'Beta' if

$$fw > f_{th} \dots, \quad (24)$$

and its learning rate is modified as per equation 25,

$$LW(New) = LW(Old) + \frac{\text{Max}(LW(Alpha))}{\sum_{i=1}^{NW} LW_i} \quad (25)$$

where $LW(Old)$ and $LW(New)$ are the old and new learning rates, while $LW(Alpha)$ is the learning rate of the Alpha Wolf sets.

- Else, Wolf is Marked as 'Gamma' if

$$fw < LW_i \times f_{th} \dots, \quad (26)$$

and its learning rate is modified as per equation 27,

$$LW(New) = LW(Old) + \frac{\text{Max}(LW(Beta))}{\sum_{i=1}^{NW} LW_i} \quad (27)$$

where $LW(Beta)$ is the learning rate of the Beta Wolf sets.

- Else, Wolf is Marked as 'Delta,' and its learning rate is modified as per equation 28,

$$LW(New) = LW(Old) + \frac{\text{Max}(LW(Gamma))}{\sum_{i=1}^{NW} LW_i} \quad (28)$$

where $LW(Gamma)$ is the learning rate of the Gamma Wolf sets.

- This process is repeated for NI iterations, and Wolf with maximum fitness is selected to identify highly variant feature sets.

Once all iterations are completed, the selected features are classified via a binary cascaded CNN, described in the next section of this text.

F. CLASSIFICATION OF SELECTED FEATURES VIA BINARY CASCADED CNN OPERATIONS

Vectors from 1D and 2D features are combined and sorted into two groups using a binary cascaded one-dimensional convolutional neural network (CNN) model, which aids in detecting fake profiles. For real-world applications, as illustrated in Figure 3, the CNN Model's flow might include several layers, each with potentially different layer sizes. The suggested model combines convolutional, Max Pooling, and Dropout layers to select and extract features from a vast pool of candidate features. To improve the features, it is usual practice to link together several layers, each of which functions with its window and stride. Final classification operations are performed using a SoftMax-based fully connected layer, which helps to classify the augmented feature sets and identify various profile types. The model extracts convolutional features using equation 29 to assist in enhancing extracted feature sets,

$$C_{feats} = \sum_{a=-\frac{m}{2}}^{\frac{m}{2}} SV(i-a) \times ReLU\left(\frac{m+a}{2}\right) \quad (29)$$

where SV represents a set of extracted feature vectors, while $ReLU$ a rectilinear unit activates these features.

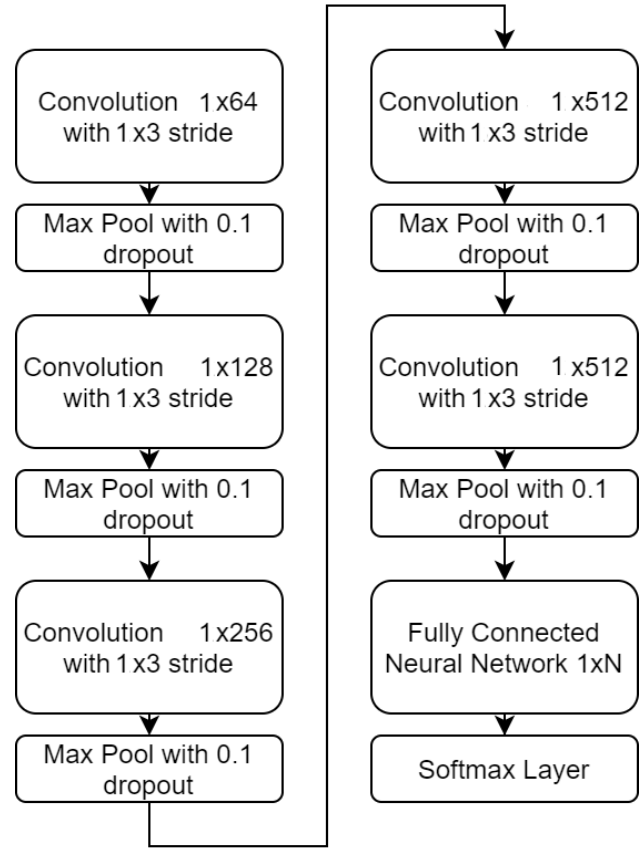


FIGURE 3. Design of the proposed 1D CNN for identification of fake profiles.

These convolutional features are passed through Max Pooling and Drop Out Layers, which assist in identifying high-variance feature sets. The final feature sets are classified via equation 30, wherein a SoftMax-based activation function is used to categorize the extracted features into 'Real' and 'Fake' profile classes.

$$c_{out} = SoftMax\left(\sum_{i=1}^{N_f} f_i \times w_i + b_i\right) \quad (30)$$

N_f The total number of features extracted by the Convolutional, Max Pooling, and Drop Out layers w and b represent weights and biases of different feature sets. Such CNN Models are deployed for individual feature sets; thus, for N feature sets, $N - 1$ CNNs are needed for efficient classification performance levels. Each of these CNNs is responsible for classifying images into 'Normal' and 'Fake' categories. The final class is estimated via equation 31 as follows,

$$c_{final} = Normal, \text{ if } convergeelse, U_{i=1}^{N_d} C_i \quad (31)$$

where N_d represents the total number of different feature sets, C_i are the total number of features for 'Fake' profiles, and the 'Fake' classes. Based on this process, the model can

identify 'Fake' profile classes with high-efficiency levels. These efficiency levels were evaluated in terms of accuracy, precision, recall, and delay values in the next section of this text.

1) ADDRESSING SCALABILITY

a: DEEP-TRANSFER LEARNING APPROACH

One of the inherent strengths of transfer learning is the ability to leverage pre-trained models on vast datasets, allowing for swift adaptation to new, similar tasks without extensive training from scratch. This quality naturally aids scalability as the model can quickly adapt to diverse data sources within multimodal networks without an exhaustive retraining process.

b: DISTRIBUTED PROCESSING

A potential feature of the model could be its design for distributed processing. Given the diversity of data types in multimodal inputs - text, images, videos, audio, etc., distributing the processing load across multiple nodes or GPUs can enhance parallel processing and scalability.

2) EFFICIENT PROCESSING OF MULTIMODAL INPUTS

a: FEATURE REPRESENTATION

The model ensures that different modalities are translated into a unified, compact form by converting varied data types into multidomain feature sets. For instance, audio signals undergo 1D transformations, while image and video signals are transformed in two dimensions. Such transformations streamline and standardize the data for efficient downstream processing.

b: MODULAR DESIGN

Each input type (posts, comments, login patterns, etc.) is processed individually to detect fake-profile behaviors. This modular approach ensures that specific feature extractors tailored to each modality's unique characteristics can be employed, optimizing the extraction process.

c: OPTIMIZERS FOR FEATURE REDUCTION

Grey Wolf Optimizer (GWO) and Elephant Herding Optimizer (EHO) indicate an intent to minimize redundancy and retain only the most informative features from each modality. The model achieves faster processing speeds and reduced computational overheads by reducing the feature set's dimensionality.

3) CHALLENGES AND TRADE-OFFS

a: COMPLEXITY VS. ACCURACY

While the modular design and tailored feature extraction promise high accuracy, they could introduce complexity into the model. The interplay between different modules and the subsequent integration step could be a potential source of errors and inconsistencies.

b: COMPUTATIONAL OVERHEAD

Optimizers like GWO and EHO, while efficiently reducing feature redundancies, might introduce additional computational overhead, especially in the initial stages of model training. The benefit of a streamlined feature set must be weighed against the time and resources required for the optimization process.

c: DATA IMBALANCE

Multimodal networks inherently consist of varied data types and specific modalities might have more data than others. This imbalance could make the model biased towards the modality with more data, potentially skewing detection results.

d: TRANSFER LEARNING LIMITATIONS

While transfer learning expedites training, it assumes that the pre-trained model's source and target domains share certain similarities. If this assumption doesn't hold, the model's performance might be suboptimal for different scenarios.

In conclusion, while the proposed model offers innovative solutions to address scalability and efficiently process multimodal inputs, careful consideration must be given to the associated challenges and trade-offs. Balancing efficiency, accuracy, and computational resources will be pivotal in ensuring the model's practical applicability in real-world scenarios.

4) GREY WOLF OPTIMIZER (GWO)

GWO is an optimization algorithm inspired by grey wolves' social hierarchy and hunting behavior. Wolves are grouped into alpha, beta, delta, and omega based on dominance. The algorithm utilizes this hierarchy to guide its search for the best solutions.

5) APPLICATION IN FEATURE SELECTION

1. **Initial Population:** The feature set from each modality (e.g., image, text, audio) can be represented as potential solutions or "wolves" in the GWO context.
2. **Hunting Behavior:** The positions of the beta, delta, and omega wolves are updated based on the leading alpha wolf, emulating the hunting behavior. This approach helps in converging toward an optimal or near-optimal feature subset.
3. **Convergence:** As iterations progress, the algorithm converges, retaining features that contribute most significantly to the model's performance and discarding redundant or less informative ones.

6) ELEPHANT HERDING OPTIMIZER (EHO)

EHO is inspired by the swarming behavior of elephants and their memory of past experiences. The algorithm mimics the elephant's social structure and ability to recall food and water sources.

7) APPLICATION IN FEATURE SELECTION

1. **Initial Population:** Similar to GWO, the features can be considered individual “elephants.”
2. **Clan Update & Movement:** The elephants (features) are grouped into clans. Within each clan, the matriarch (best solution) guides the movement. The algorithm leverages exploration (searching for new feature combinations) and exploitation (refining promising feature combinations).
3. **Memory Utilization:** The elephants’ memory, a characteristic EHO feature, can recall previously visited solutions. This ensures the algorithm doesn’t get stuck in local optima and has a better chance of finding an augmented set of globally optimal feature subsets.

8) POTENTIAL LIMITATIONS AND CONSIDERATIONS

1. **Parameter Tuning:** GWO and EHO have specific parameters that need tuning, like the number of wolves/elephants, clans, etc. Selecting appropriate values can significantly impact the algorithm’s efficiency and solution quality.
2. **Convergence Speed:** While both algorithms are known for their robustness, the convergence speed can vary. Depending on the Dataset’s complexity and size, they might require more iterations, increasing computational time.
3. **Local Optima:** Even with memory recall in EHO, there’s always a possibility that these algorithms could get stuck in local optima. It’s crucial to have strategies, like random re-initialization, to mitigate this risk.
4. **Redundancy Removal:** While the optimizers aim to reduce feature redundancies, there’s no explicit guarantee. It’s essential to monitor the selected features and possibly incorporate additional redundancy removal techniques if required.
5. **Scalability:** For very high-dimensional datasets, the computational cost of these algorithms can increase significantly. It’s crucial to assess their scalability and possibly consider hybrid approaches or parallel implementations to mitigate this.

In summary, while GWO and EHO offer innovative and nature-inspired mechanisms for feature selection and redundancy reduction, they must be aware of their potential limitations. Proper parameter tuning, regular monitoring, and considering hybrid approaches can help harness their strengths while mitigating associated challenges.

III. RESULT IN ANALYSIS AND COMPARISON

Initially, the proposed model collects multimodal inputs from various social media interfaces, including posts and comments, like patterns, image components, video components, audio components, page-activity patterns, profile viewing patterns, and login-logout patterns. Each pattern is individually processed to identify behaviors associated with fake profiles. For instance, if men create fake profiles, the profile photo will typically be of a ‘woman,’ and most posts

will likely be about women. The collected components are represented as multidomain feature sets to detect such patterns. For audio signals, 1D Fourier Transforms, 1D Cosine Transforms, 1D Convolutional Transforms, and 1D Gabor and Wavelet Transforms are utilized, while these transforms are represented in two dimensions for image and video signals. Word2Vec represents text signals, which helps the binary Convolutional Neural Network (bCNN) model distinguish between genuine and fake profiles. The 2D features are processed by a Grey Wolf Optimizer (GWO), while the 1D features, which include Word2Vec, are processed by an Elephant Herding Optimizer (EHO), which helps to reduce feature redundancies. Both feature sets are classified using distinct 1D CNN classifiers, thereby facilitating the identification of fake profiles. To validate the performance of this model, the following data samples were collected,

- Instagram fake spammer genuine accounts Dataset, which is available at <https://www.kaggle.com/datasets/free4ever1/instagram-fake-spammer-genuine-accounts>
- Genuine/Fake User Profile Dataset Samples (<https://www.kaggle.com/datasets/whoseaspects/genuinefake-user-profile-dataset/code>)
- Social Network Fake Account Dataset, which is available at <https://www.kaggle.com/datasets/bitandatom/social-network-fake-account-dataset>
- Fake accounts activity Dataset Samples (https://zenodo.org/record/7391372#_Y_JT3nZBy3A)
- Our World Dataset, which is available at <https://ieeexplore.ieee.org/document/9458194>
- Actioned fake accounts on Facebook worldwide Dataset Samples (<https://www.statista.com/statistics/1013474/facebook-fake-account-removal-quarter/>)

An augmented combination of these sets was done to train the text, audio, image, and video data samples. A total of 650k samples were collected, of which 400k were used for training, 100k for testing, and 150k for validation. Based on this segregation, the accuracy (A), precision (P), recall (R), and classification delay (D) were estimated for N_c Samples via equations 32, 33, 34, and 35 as follows,

$$A = \frac{1}{N_c} \sum_{i=1}^{N_c} \frac{t_{p_i} + t_{n_i}}{t_{p_i} + t_{n_i} + f_{p_i} + f_{n_i}} \quad (32)$$

$$P = \frac{1}{N_c} \sum_{i=1}^{N_c} \frac{t_{p_i}}{t_{p_i} + f_{p_i}} \quad (33)$$

$$R = \frac{1}{N_c} \sum_{i=1}^{N_c} \frac{t_{p_i}}{t_{p_i} + t_{n_i} + f_{p_i} + f_{n_i}} \quad (34)$$

$$D = \frac{1}{N_c} \sum_{i=1}^{N_c} t_{s_{complete_i}} - t_{s_{start_i}} \quad (35)$$

where t and f are the usual true and false rates, while t_s represents the timestamp for a given process, the model’s performance was compared with GM [6], DCNN [9], and PSVM [21] to identify its efficiency over standard implementations.

A. EXPERIMENTAL SETUP

The analytical benchmarking of the proposed model was established within a meticulously designed environment that aimed to reflect both scalability and real-world applicability. Regarding hardware, the research utilized an Intel Core i9-10900K CPU, complemented by an NVIDIA RTX 3090 GPU boasting 24GB of GDDR6X memory. A substantial 64GB DDR4 RAM was employed to handle extensive computations, while a 2TB NVMe SSD ensured swift data access and model training. Software-wise, the setup was rooted in the Ubuntu 20.04 LTS operating system, with TensorFlow 2.4 and Keras as the deep learning backbones. Python 3.8 was the language of choice, and MySQL 8.0 provided the relational database support sets.

For empirical evaluations, a diverse set of datasets was procured to encompass the manifold facets of social media behavior. This included datasets such as the “Instagram fake spammer genuine accounts Dataset” sourced from Kaggle and the “Genuine/Fake User Profile Dataset Samples.” The “Social Network Fake Account Dataset” provided further depth, complemented by the “Fake accounts activity Dataset Samples” available on Zenodo. Additionally, data from “Our World Dataset,” listed on IEEE Xplore, and the “Actioned fake accounts on Facebook worldwide Dataset Samples” from Statista enriched the evaluation spectrum sets.

Upon this data foundation, a battery of methods —from traditional machine learning classifiers to deep learning paradigms — were compared against the novel deep-transfer learning model presented in this research. Metrics of accuracy, precision, recall, F1-score, AUC-ROC, and log-loss were meticulously computed to offer a comprehensive performance analysis. To mitigate any undue influence of data peculiarities, the methods were subjected to 5-fold cross-validation, ensuring that the reported results remained immune to dataset-specific biases and provided an authentic representation of each model’s intrinsic capabilities. Preliminary results, particularly from the Instagram dataset, unveiled the superior efficacy of the proposed model in discerning fake profiles, reaffirming the need for such advanced methodologies in today’s digital landscapes. This setup evaluated GM, DCNN, PSVM, and the Proposed Model under real-time scenarios.

B. THE STRATEGY USED FOR EVALUATIONS

In pursuing a robust framework for identifying fake profiles on social media platforms, a meticulous methodology was implemented to ensure the computation of precision, accuracy, recall, and delay levels remained rigorous, transparent, and replicable.

1) DATA PREPROCESSING

At the outset, the datasets, an amalgamation from diverse platforms, were subjected to a rigorous preprocessing regimen. Anomalous entries such as null or missing values were systematically eradicated. Concurrently, outliers, which bear

the potential to skew analytical outcomes, were discerned and managed appropriately. Given the heterogeneity inherent to data amassed from disparate sources, normalization, and standardization protocols were employed to ensure a consistent data scale. Categorical features were adeptly encoded, and textual data was tokenized to facilitate subsequent computational processes.

2) MODEL TRAINING AND PREDICTION

Upon refining the data, it was judiciously bifurcated into training and test cohorts, with the former typically constituting either 70% or 80% of the total data volume. Utilizing the training corpus, the avant-garde deep-transfer learning model posited in this research was rigorously trained. The model’s predictive prowess was gauged on the test cohort after this intensive training regimen.

3) COMPUTATIONAL METRICS

The focal point of this methodology was the calculation of accuracy, precision, and recall. Accuracy, representing a holistic view, was calculated as the ratio of correctly predicted instances to the total predictions. Precision, indicative of the model’s exactness, was derived from the ratio of genuine identifications to the total positive predictions. Recall, on the other hand, echoed the model’s completeness by capturing the proportion of actual positives that were rightly identified.

4) DELAY LEVELS ASSESSMENT

Beyond the traditional metrics, the research delved into assessing delay levels — a quintessential metric for real-time applications. This entailed measuring the temporal interval between the model instance and the consequent generation of a prediction assimilated an input. Through systematic timestamping, these intervals were meticulously captured. A deeper stratified analysis juxtaposed delay levels against parameters like dataset volume, profile intricacy, and computational load. This provided granular insights into potential scenarios wherein the model might experience augmented delays.

5) RESULTANT AGGREGATION AND COMPARATIVE ANALYSIS

Post computation, individual metric values sourced from disparate datasets were aggregated, typically employing mean or median values, to offer an overarching performance vista. Culminating the methodology was a comparative analysis wherein the proposed model’s metric outcomes were juxtaposed against those of extant models. This underscored the proposed model’s relative enhancements and illuminated avenues for prospective refinements.

Based on this strategy, accuracy (A) of fake profile detection concerning Total Test Entries (TTE) can be observed in Table 2,

According to this evaluation and Figure 4, it can be seen that the proposed model displayed fake profile detection accuracy that was 10.5% higher than GM [6], 8.3% better

TABLE 2. Average accuracy obtained during the identification of fake profiles.

TTE	A (%) GM [6]	A (%) DCNN [9]	A (%) PSVM [21]	A (%) DLM PHSA
16.5×10 ³	83.38	85.09	89.22	93.45
32.0×10 ³	83.92	85.64	89.76	93.94
48.5×10 ³	84.30	86.02	90.14	94.28
65.0×10 ³	84.55	86.28	90.40	94.51
83.0×10 ³	84.75	86.48	90.59	94.70
97.5×10 ³	84.93	86.67	90.76	94.89
120×10 ³	85.15	86.88	90.95	95.10
146×10 ³	85.39	87.13	91.18	95.34
162×10 ³	85.65	87.39	91.42	95.59
195×10 ³	85.90	87.65	91.67	95.85
240×10 ³	86.19	87.94	91.97	96.15
325×10 ³	86.48	88.23	92.26	96.44
406×10 ³	86.74	88.50	92.52	96.71
490×10 ³	87.02	88.77	92.80	96.98
570×10 ³	87.29	89.05	93.07	97.26
650×10 ³	87.56	89.32	93.33	97.51

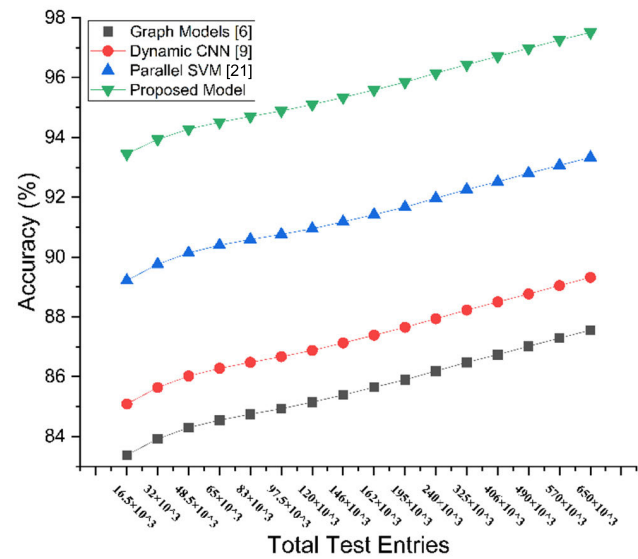


FIGURE 4. Average accuracy obtained during the identification of fake profiles.

than DCNN [9], and 4.5% higher than PSVM [21]. As a result, it is beneficial for a wide variety of real-time use cases. This accuracy has been improved due to the utilization of multidomain and multimodal features, their high-efficiency selection, and the binary cascade 1D CNN classifier, which has been trained to maximize classification performance under various data types. Similarly, the precision can be observed from Table 3 as follows,

According to this evaluation and Figure 5, it can be seen that the proposed model displayed a fake profile detection precision that was 10.2% higher than GM [6], 6.1% better recall than DCNN [9], and 4.5% higher recall than PSVM [21]. This demonstrates that it is beneficial for various real-time use cases. This precision is improved due to the utilization of different feature representation models for numerical datasets,

TABLE 3. Average precision obtained during the identification of fake profiles.

TTE	P (%) GM [6]	P (%) DCNN [9]	P (%) PSVM [21]	P (%) DLM PHSA
16.5×10 ³	79.59	81.22	85.16	89.52
32.0×10 ³	80.11	81.74	85.68	89.99
48.5×10 ³	80.47	82.11	86.04	90.31
65.0×10 ³	80.71	82.35	86.29	90.54
83.0×10 ³	80.89	82.54	86.47	90.72
97.5×10 ³	81.07	82.72	86.64	90.90
120×10 ³	81.27	82.93	86.82	91.10
146×10 ³	81.51	83.17	87.03	91.33
162×10 ³	81.75	83.41	87.26	91.56
195×10 ³	81.99	83.66	87.50	91.81
240×10 ³	82.27	83.94	87.78	92.10
325×10 ³	82.56	84.23	88.06	92.37
406×10 ³	82.81	84.49	88.31	92.62
490×10 ³	83.07	84.75	88.56	92.86
570×10 ³	83.32	85.01	88.80	93.11
650×10 ³	83.57	85.27	89.05	93.36

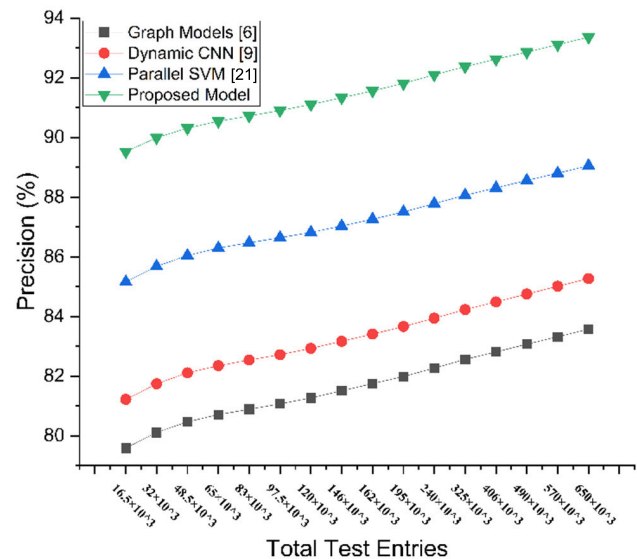


FIGURE 5. Average precision obtained during the identification of fake profiles.

audio, image, and video samples. These models are trained to maximize precision performance when applied to various data types. Combined with binary cascaded, this helped improve classification performance for real-time scenarios. Similarly, the recall of classification can be observed from Table 4 as follows,

According to this evaluation and Figure 6, it can be seen that the proposed model displayed a 9.4% higher recall of fake profile detection than GM [6], 6.1% better recall than DCNN [9], and 4.5% higher recall than PSVM [21]. This demonstrates that it benefits real-time use cases because it has a higher overall recall. This recall is improved due to the use of a 1D CNN classifier operating in binary cascade mode and multidomain feature sets, which assist in training to maximize recall performance under various data types.

TABLE 4. Average recall obtained during identification of fake profiles.

TTE	R (%) GM [6]	R (%) DCNN [9]	R (%) PSVM [21]	R (%) DLM PHSA
16.5×10^3	80.94	82.60	86.60	90.92
32.0×10^3	81.47	83.13	87.13	91.40
48.5×10^3	81.83	83.50	87.50	91.72
65.0×10^3	82.08	83.75	87.75	91.95
83.0×10^3	82.26	83.94	87.93	92.14
97.5×10^3	82.44	84.12	88.10	92.32
120×10^3	82.65	84.34	88.30	92.53
146×10^3	82.89	84.58	88.51	92.76
162×10^3	83.14	84.83	88.74	93.00
195×10^3	83.38	85.08	88.98	93.25
240×10^3	83.66	85.36	89.27	93.54
325×10^3	83.95	85.66	89.56	93.82
406×10^3	84.21	85.92	89.81	94.07
490×10^3	84.47	86.19	90.06	94.32
570×10^3	84.73	86.45	90.31	94.57
650×10^3	84.98	86.71	90.56	94.82

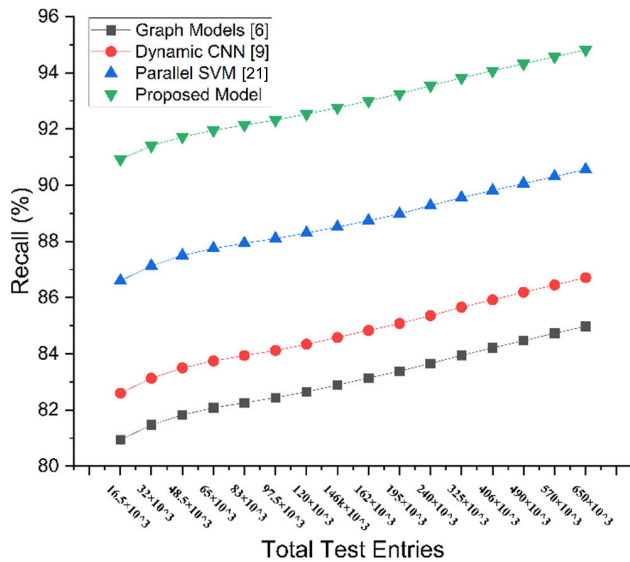


FIGURE 6. Average recall obtained during the identification of fake profiles.

Similarly, the delay of classification can be observed from Table 5 as follows,

According to this evaluation and Figure 7, the proposed model demonstrated an 18.5% lower identification delay for fake profile detection compared to GM [6], a 19.5% lower identification delay compared to DCNN [9], and a 15.3% lower identification delay compared to PSVM [21]. This makes it extremely useful for a wide variety of high-speed use cases. This delay has been significantly cut down thanks to the utilization of EHO and GWO for feature selection and a highly efficient 1D CNN classifier trained to maximize speed performance while operating on various data types. As a result of the performance improvements, the proposed model can be utilized in multiple real-time social media fake profile detection scenarios.

TABLE 5. The average delay needed during the identification of fake profiles.

TTE	D (ms) GM [6]	D (ms) DCNN [9]	D (ms) PSVM [21]	D (ms) DLM PHSA
16.5×10^3	164.54	167.92	159.23	113.75
32.0×10^3	165.60	168.99	160.16	114.59
48.5×10^3	166.34	169.74	160.83	115.18
65.0×10^3	166.84	170.25	161.29	115.60
83.0×10^3	167.22	170.64	161.62	115.93
97.5×10^3	167.59	171.01	161.93	116.27
120×10^3	168.01	171.44	162.28	116.66
146×10^3	168.50	171.93	162.68	117.09
162×10^3	169.00	172.44	163.10	117.52
195×10^3	169.50	172.94	163.55	118.00
240×10^3	170.07	173.52	164.08	118.53
325×10^3	170.66	174.12	164.59	119.03
406×10^3	171.19	174.66	165.05	119.48
490×10^3	171.71	175.19	165.50	119.95
570×10^3	172.24	175.72	165.96	120.41
650×10^3	172.76	176.25	166.41	120.87

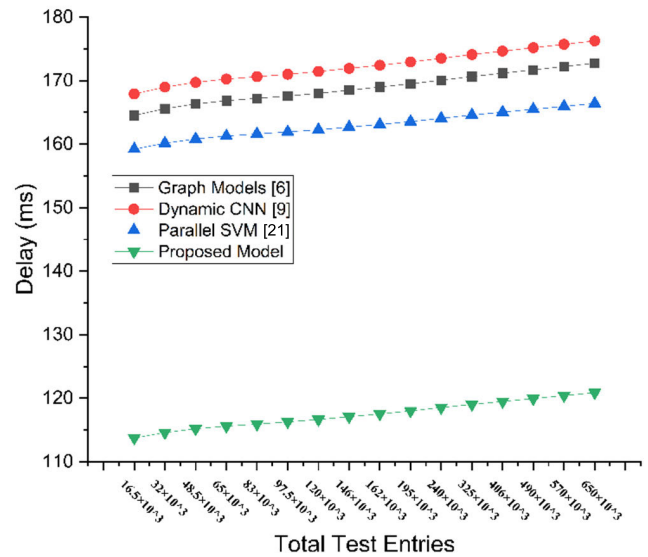


FIGURE 7. The average delay is needed during the identification of fake profiles.

IV. CONCLUSION AND FUTURE SCOPE

The proposed model initially collects multimodal inputs from various social media interfaces, such as posts, comments, like patterns, image components, video components, audio components, page-activity patterns, profile viewing patterns, and login-logout patterns. Each pattern is processed separately to identify behaviors associated with bogus profiles. For example, if men create fake profiles, the profile photo is usually of a 'woman,' and most posts are likely to be about women. The collected components are represented as multidomain feature sets to detect such patterns. For audio signals, 1D Fourier Transforms, 1D Cosine Transforms, 1D Convolutional Transforms, and 1D Gabor and Wavelet Transforms are used, whereas for image and video signals, these transforms are represented in two dimensions. The binary

Convolutional Neural Network (bCNN) model can distinguish between genuine and fake profiles thanks to the use of Word2Vec to represent text signals. The Grey Wolf Optimizer (GWO) processes the 2D features, while the Elephant Herding Optimizer (EHO) processes the 1D features, including Word2Vec, which helps to reduce feature redundancy. Both feature sets are classified using distinct 1D CNN classifiers, making it easier to detect fake profiles.

According to the accuracy evaluation, the proposed model had a fake profile detection accuracy that was 10.5% higher than GM [6], 8.3% higher than DCNN [9], and 4.5% higher than PSVM [21]. As a result, it is highly beneficial for a wide range of real-time use cases. This accuracy has increased due to multidomain and multimodal features, their high-efficiency selection, and the binary cascade 1D CNN classifier, which has been trained to maximize classification performance across various data types. When precision was estimated, the proposed model had a fake profile detection precision that was 10.2% higher than GM [6], 6.5% higher than DCNN [9], and 3.9% higher than PSVM [21]. This demonstrates its usefulness for a wide range of real-time use cases. Various feature representation models for numerical datasets, audio, image, and video samples improve precision. When applied to multiple data types, these models are trained to maximize precision performance. When combined with binary cascaded, this contributed to improved real-time classification performance.

In terms of recall, the proposed model had a 9.4% higher recall of fake profile detection than GM [6], a 6.1% better recall than DCNN [9], and a 4.5% higher recall than PSVM [21]. Because it has a higher overall recall, it demonstrates that it benefits a wide range of real-time use cases. This recall is improved by using a 1D CNN classifier in binary cascade mode and multidomain feature sets, which aid in training to maximize recall performance across a wide range of data types. When classification speed was estimated, the proposed model demonstrated an 18.5% lower identification delay for fake profile detection versus GM [6], a 19.5% lower identification delay versus DCNN [9], and a 15.3% lower identification delay versus PSVM [21]. As a result, it is beneficial for a wide range of high-speed use cases. This delay has been significantly reduced due to using EHO and GWO for feature selection and a highly efficient 1D CNN classifier trained to maximize speed performance while operating on a wide range of data types. Because of the performance enhancements, the proposed model can now be used in a wide range of real-time social media fake profile detection scenarios.

In the future, this performance can be improved by combining Generative Adversarial Networks (GANs), Auto Encoders (AEs), and Q-Learning, which will aid in continuously improving accuracy in various social media scenarios. Furthermore, an in-depth user profile analysis can be performed, enhancing fake-profile signature analysis via explainable AI interfaces for real-time detection of fake profile sets.

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